A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Arenes

Benzene has six carbons and three apparent double bonds, yet it refuses to react like an alkene. This topic explains why: its electrons are spread around the whole ring, which makes it unusually stable and steers it towards substitution rather than addition. It also covers the reactions that put new groups onto the ring, and why phenol reacts so much more readily.

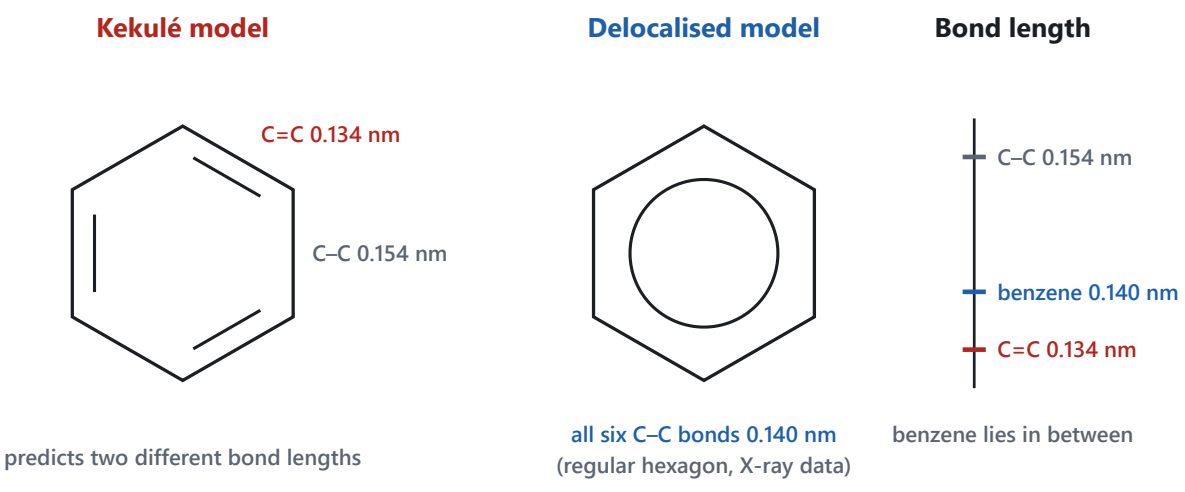
SPEC 18.1 Evidence for the structure and stability of benzene

WHAT THIS SPEC POINT ASKS

Use enthalpy (thermochemical) data, X-ray diffraction data and infrared data as evidence for the structure of the benzene ring and for its extra stability.

Two models of benzene

- Benzene is C_6H_6: six carbons in a flat ring, each carbon bonded to one hydrogen.
- **Kekulé model**: a ring of alternating C=C double and C–C single bonds (cyclohexa-1,3,5-triene). If this were right, benzene would behave like an alkene with three separate double bonds.
- **Delocalised model**: all six C–C bonds are identical. Six π electrons are spread over the whole ring (18.2). You can draw benzene either way in equations and mechanisms, but the evidence below supports the delocalised model.

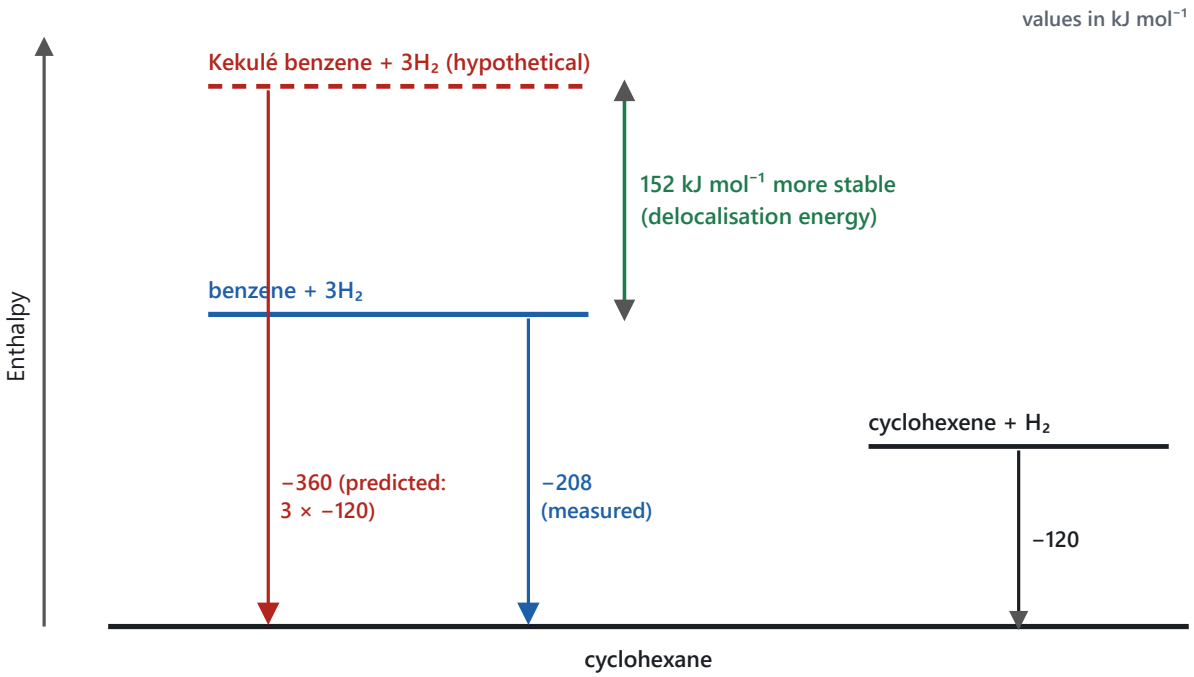


The Kekulé structure predicts two bond lengths. X-ray diffraction finds one: 0.140 nm, between C–C and C=C.

Every answer on this spec point has the same shape: **"Kekulé predicts X, the measurement shows Y, so the bonds are delocalised."**

Evidence 1: enthalpy change of hydrogenation

- Cyclohexene (one C=C) + H_2 → cyclohexane: ΔH = **−120 kJ mol⁻¹**.
- A Kekulé benzene has three C=C bonds, so it should give 3 × (−120) = **−360 kJ mol⁻¹**.
- The measured value for benzene is only **−208 kJ mol⁻¹**: **152 kJ mol⁻¹ less exothermic** than predicted.
- So less energy is released, which means benzene started lower in energy: it is **more stable** than the Kekulé structure. This extra stability is the **delocalisation energy** (also called resonance energy or stabilisation energy).



Real benzene sits 152 kJ mol⁻¹ below the hypothetical Kekulé structure, so it releases less energy when hydrogenated.

WORKED EXAMPLE

Worked example: stabilisation from hydrogenation data

Data: cyclohexene → cyclohexane, ΔH = −118 kJ mol⁻¹; benzene → cyclohexane, ΔH = −205 kJ mol⁻¹.

Kekulé prediction = 3 × (−118) = −354 kJ mol⁻¹.

Difference = −205 − (−354) = **+149 kJ mol⁻¹**.

Benzene's hydrogenation is 149 kJ mol⁻¹ **less exothermic** than expected, so benzene is 149 kJ mol⁻¹ more stable than the Kekulé structure, because its π electrons are delocalised.

Papers use slightly different data (−118/−205 or −120/−208). Always use the numbers in the question.

WORKED EXAMPLE

Worked example: the same idea from enthalpies of combustion

ΔcH: cyclohexene −3752, cyclohexa-1,4-diene −3584, benzene −3267 kJ mol^{-1}.

1. Each extra C=C makes ΔcH less negative, because there are two fewer H atoms to burn. Cyclohexene − diene: −3752 − (−3584) = **−168 kJ mol^{-1}**, so each extra C=C is worth 168 kJ mol^{-1}.
2. Predicted ΔcH for a Kekulé triene = −3584 − (−168) = **−3416 kJ mol^{-1}**.
3. Benzene's real value is −3267, so it releases 3416 − 3267 = **149 kJ mol^{-1} less** energy on combustion.
4. Benzene is 149 kJ mol^{-1} more stable than the Kekulé structure, because of delocalisation (WCH15 Jun 2021).

Evidence 2: X-ray diffraction (bond lengths)

- X-ray diffraction maps the electron density, so it gives bond lengths.
- Kekulé predicts **two** lengths: three short C=C (0.134 nm) and three long C–C (0.154 nm), giving an irregular hexagon.
- Benzene has **six equal** C–C bonds of **0.140 nm**: a regular hexagon. The length is **in between** a single and a double bond.

Evidence 3: infrared spectrum

- An alkene C=C stretch absorbs at about **1669–1645 cm^{-1}** (Data Booklet). Benzene shows **no** absorption there.
- Instead it has the arene peaks at about **1600, 1580, 1500 and 1450 cm^{-1}** and an arene C–H stretch near **3030 cm^{-1}**. These show a different kind of carbon–carbon bond from an alkene's.

Evidence 4: chemical behaviour

- Benzene does **not decolourise bromine water**, and it needs a catalyst to react with bromine at all. It reacts by **substitution**, not addition. An alkene with localised C=C bonds would add bromine at once (see 18.3).

MARK-SCHEME WORDING

- Thermochemistry: Kekulé expected "3 × −118 = −354" (or 3 × −120 = −360); benzene is "149 / 152 kJ mol^{-1} less exothermic than expected" or "more stable than expected"; reason "the π electrons are delocalised". "Delocalisation energy" or "resonance stability" is accepted.
- Use comparative words: "less exothermic", "less negative", "more stable". "Higher" or "lower" on their own are not accepted.
- X-ray: "bond lengths in benzene are **equal**", "in between C=C and C–C" or "longer than C=C in cyclohexene"; and say what Kekulé predicts: "Kekulé structure would have shorter double bonds and

longer single bonds". "Regular hexagon" is accepted; "bond lengths are similar" is not.
- IR: specific arene wavenumbers, and no C=C alkene peak.

Often asked as: the 6-mark "evidence for benzene's structure" (WCH15 Jun 2023), a hydrogenation or combustion calculation, explain X-ray data, MCQs on bond lengths.

COMMON MISTAKE

Comparing benzene's −208 with cyclohexene's −120. The comparison is with the **hypothetical Kekulé triene** (−360).

Drawing the enthalpy-level diagram as a reaction profile with a hump. It is a set of horizontal levels with vertical arrows.

Saying X-ray shows "the bonds are the same" without saying what Kekulé would show. The "Kekulé double bonds would be shorter" half-mark is the one most often missed (Jan 2024 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Isomer count**: a Kekulé structure would give **two** isomers of 1,2-dichlorobenzene (Cl atoms across a C=C or across a C–C). Only one exists (WCH15 Jan 2024).
- **Conjugated dienes**: cyclohexa-1,3-diene is only about 8 kJ mol^{-1} more stable than 2 × (−120) predicts, because its two C=C bonds are conjugated but not a closed ring. Benzene's continuous ring of delocalisation is much more effective (WCH15 Jan 2026).
- **Bond enthalpy cycle**: the mean aromatic C–C bond enthalpy (about 507 kJ mol^{-1}) comes out higher than the average of C–C and C=C (about 480), again because of delocalisation (WCH15 Jun 2025).
- **Dewar and Ladenburg benzene**: other proposed C_6H_6 structures. Bromine water or NMR shows they are not benzene: benzene's H atoms all appear at about 7.3 ppm (WCH15 Oct 2021).

SPEC 18.2 The delocalised model: overlap of p orbitals

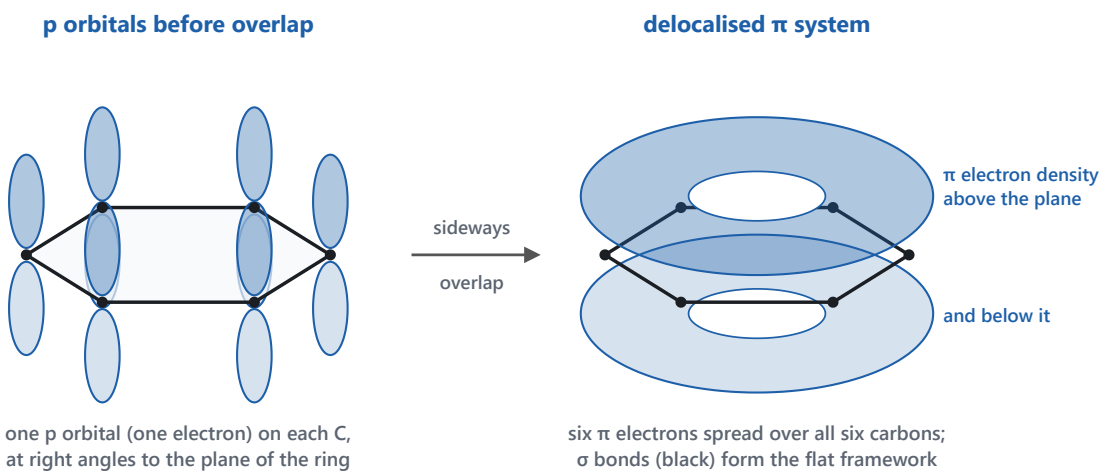
WHAT THIS SPEC POINT ASKS

Understand that the delocalised model of benzene involves sideways overlap of p orbitals to form π bonds.

Building the model

1. Each carbon uses three of its four outer electrons to form **σ bonds**: two to neighbouring carbons and one to a hydrogen. This gives a **flat** (planar) ring with bond angles of **120°**.
2. Each carbon has **one electron left in a p orbital**. The p orbital sticks out **above and below** the plane of the ring, at right angles to it.

3. Each p orbital **overlaps sideways** with the p orbitals on **both** neighbouring carbons, not just one.
4. The result is a continuous **ring of π electron density above and below the plane**, holding **six delocalised π electrons** shared by all six carbons.



Left: one p orbital on each carbon. Right: sideways overlap all round the ring gives a delocalised π system above and below the plane.

- Because every carbon shares in the same π system, all C–C bonds are the same length (0.140 nm).
- Delocalisation spreads the electrons over a bigger region, which lowers the energy. That is the source of the extra stability in 18.1.
- The circle in the hexagon stands for the six delocalised electrons.

EXAM TIP

In an alkene, the p orbitals on **two** carbons overlap to make one **localised** π bond. In benzene, the p orbitals on **six** carbons overlap all round the ring. Use that contrast whenever 18.2 and 18.3 come up together.

MARK-SCHEME WORDING

- "six σ bonds between carbon atoms" (the flat framework), "six p (p_z) orbitals not involved in σ bonding", "overlap (continuously) **above and below** the ring" to form "a (large) π system". All marks can come from a labelled diagram.
- Hybridisation language ("sp^2 orbitals overlap head-on") is accepted although not on the spec. "π orbitals" is not accepted.

Often asked as: describe and explain the bonding (2–3 marks, WCH15 Jan 2024, Jun 2025); MCQs on sideways overlap and on counting bonds.

COMMON MISTAKE

Unlabelled diagrams, and leaving out the σ framework. Say what the σ bonds are as well as the π system (Jan 2024 report).

Saying "p orbitals overlap end-on". End-on overlap makes σ bonds; π bonds come from **sideways** overlap.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Counting bonds: benzene has **12 σ bonds** (6 C–C + 6 C–H). Drawn as Kekulé it has **3 π bonds** (WCH15 Jun 2022 MCQ).

SPEC 18.3 Why benzene resists bromination

WHAT THIS SPEC POINT ASKS

Explain why benzene reacts with bromine much less readily than an alkene does: delocalised π electrons in benzene, localised π electron density in an alkene.

Alkene: localised π bond

- In cyclohexene, the two π electrons sit **between just two carbons**, so the electron density there is **high**.
- When a Br_2 molecule approaches, this concentrated negative charge **repels the electrons in Br–Br** and **induces a dipole** (Brδ+–Brδ−).
- The π electrons attack the δ+ bromine. The alkene adds bromine at room temperature: **electrophilic addition**. Bromine water is decolourised at once.

Benzene: delocalised π system

- Benzene's six π electrons are spread over **six carbons**, so the electron density between any two carbons is **lower**.
- The ring **cannot polarise** Br_2 enough to start a reaction. Benzene doesn't react with bromine water.
- Addition would also **destroy the delocalised ring** and lose about 150 kJ mol^{-1} of stability. So the **activation energy is high** and benzene is **kinetically stable**.
- With a **halogen carrier** ($FeBr_3$) to make a much stronger electrophile, Br^+, benzene does react. It undergoes **electrophilic substitution**: an H atom is replaced and the stable ring is kept (18.4, 18.5).

KEY IDEA

Model answer (benzene v cyclohexene with bromine):

- **Both** react with electrophiles because both have π electrons.
- Cyclohexene has **localised** π electron density in one C=C bond, high enough to **polarise** Br_2. It reacts at room temperature with no catalyst by **electrophilic addition** (product 1,2-dibromocyclohexane).
- Benzene's π electrons are **delocalised** over the ring, so the electron density is **lower** and it cannot polarise Br_2. It needs a **catalyst** ($FeBr_3$) and reacts by **electrophilic substitution** (bromobenzene + HBr), because substitution **keeps the stable delocalised ring**.

MARK-SCHEME WORDING

Indicative points from the 6-mark "cyclohexene, benzene and phenol with bromine" (WCH15 Oct 2022, Oct 2023, Jan 2025):

- "all are attacked by / react with electrophiles";
- "cyclohexene undergoes addition", "benzene and phenol undergo substitution";
- "cyclohexene and phenol react with (aqueous) bromine without a catalyst; benzene needs a catalyst";
- "benzene has delocalised electrons and is (kinetically) stable, so the reaction has a high activation energy";
- "cyclohexene has localised electron density in one π bond";
- phenol: "lone pair on the oxygen overlaps with the ring, increasing electron density" (18.6).

A similarity or a difference given for one compound only, with no comparison, loses the reasoning mark. "Nucleophilic" is never accepted.

Often asked as: a 6-mark compare-and-contrast, or a 4-mark "compare the reactions of benzene and cyclohexene with bromine" (WCH15 Jun 2021).

COMMON MISTAKE

"Benzene doesn't decolourise bromine, but cyclohexene does" is an observation, not an explanation. Give the electron-density reason too.

Leaving out the word **electrophilic** ("benzene undergoes substitution").

Writing cyclohexane or cyclohexa-1,3,5-triene when you mean cyclohexene. Giving dibromobenzene as the product of benzene.

Writing out full mechanisms in a 6-mark answer that didn't ask for them: it wastes time and any error costs a mark (Oct 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Kinetic v thermodynamic stability: the delocalised system makes the activation energy for forming the intermediate high, so benzene is kinetically stable (WCH15 Jun 2021 MCQ). And **HBr** is a by-product of substitution, but not of addition (WCH15 Jan 2025).

SPEC 18.4 Reactions of benzene

WHAT THIS SPEC POINT ASKS

Know these reactions of benzene: (i) combustion in air (smoky flame); (ii) bromine with a catalyst; (iii) a mixture of concentrated nitric and sulfuric acids; (iv) fuming sulfuric acid; (v) halogenoalkanes and acyl chlorides with aluminium chloride (Friedel–Crafts). Know the common uses of the products.

(i) Combustion

$2C_6H_6(l) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(l)$

- Benzene burns with a **very smoky (sooty), yellow flame**.
- Its **C : H ratio is high** (1 : 1), so there isn't enough oxygen in air for complete combustion. Unburnt **carbon** (soot) forms.
- A sooty flame is a quick test for an **arene** in an unknown (WCH15 Jun 2022).

(ii) Halogenation (bromination)

$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$

- **Reagent**: bromine (liquid). **Catalyst** (halogen carrier): **$FeBr_3$** (or iron filings, which react with Br_2 to make $FeBr_3$), or $AlBr_3$. Room temperature, dry (anhydrous) conditions.
- **Observations**: the orange-brown colour fades; **steamy fumes of HBr** are given off.
- Chlorination works the same way with Cl_2 and $AlCl_3$ or $FeCl_3$, giving **chlorobenzene**.
- **Product: bromobenzene.** It is a route into Grignard chemistry (C_6H_5MgBr, 20.2).

(iii) Nitration

$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$

- **Reagents**: **concentrated nitric acid** and **concentrated sulfuric acid** (the "nitrating mixture"). The sulfuric acid is a catalyst that generates the electrophile NO_2^+.
- **Conditions**: warm at about **50–55 °C**. Keep it **below about 55 °C** to stop further substitution (dinitrobenzene). Add the benzene slowly and cool the flask, because the reaction is exothermic.

- **Product: nitrobenzene**, a pale yellow oily liquid.
- **Uses**: reduced to **phenylamine** (19.5), which is used to make **dyes** (19.6) and drugs. Multiple nitration of methylbenzene gives the explosive **TNT**.

(iv) Sulfonation

$C_6H_6 + SO_3 \rightarrow C_6H_5SO_3H$ (or $C_6H_6 + H_2SO_4 \rightarrow C_6H_5SO_3H + H_2O$)

- **Reagent**: **fuming sulfuric acid** (oleum: concentrated H_2SO_4 with dissolved SO_3). Warm, about 40 °C. The electrophile is SO_3 (or HSO_3^+).
- **Product: benzenesulfonic acid**. The $–SO_3H$ group is joined to the ring through **sulfur**.
- **Uses**: alkylbenzenesulfonic acids (as their sodium salts) are **detergents**. Sulfonate groups also make **dyes soluble in water** (WCH15 Jun 2025).

(v) Friedel–Crafts reactions

Both need **anhydrous aluminium chloride**, $AlCl_3$, as catalyst (a halogen carrier). Heat under reflux in dry conditions, because water destroys $AlCl_3$.

	Alkylation	Acylation
Reagent	**halogenoalkane**, e.g. chloromethane CH_3Cl	**acyl chloride**, e.g. ethanoyl chloride CH_3COCl
Equation	$C_6H_6 + CH_3Cl \rightarrow C_6H_5CH_3 + HCl$	$C_6H_6 + CH_3COCl \rightarrow C_6H_5COCH_3 + HCl$
Product	**methylbenzene** (an alkylbenzene)	**phenylethanone** (an aromatic ketone)
Electrophile	CH_3^+ (a carbocation)	CH_3CO^+ (an acylium ion)

- Friedel–Crafts is the way to **make a new C–C bond to a benzene ring**.
- **Uses**: ethylbenzene (from chloroethane) is turned into phenylethene, the monomer for poly(phenylethene) (polystyrene). The ketone from acylation can be reduced to an alcohol with $NaBH_4$.

Summary table

Reaction	Reagents and conditions	Electrophile	Organic product	By-product
Bromination	Br_2, $FeBr_3$ catalyst, room temp.	Br^+	bromobenzene	HBr
Nitration	conc. HNO_3 + conc. H_2SO_4, 50–55 °C	NO_2^+	nitrobenzene	H_2O

Reaction	Reagents and conditions	Electrophile	Organic product	By-product
Sulfonation	fuming H_2SO_4, warm	SO_3	benzenesulfonic acid	(none, with SO_3)
Alkylation	RCl + anhydrous $AlCl_3$, reflux	R^+	alkylbenzene	HCl
Acylation	RCOCl + anhydrous $AlCl_3$, reflux	RCO^+	phenyl ketone	HCl

All except combustion are **electrophilic substitutions**: one H on the ring is swapped for another group, and the delocalised ring survives.

MARK-SCHEME WORDING

- Nitration: "**concentrated** sulfuric acid" (dilute is not accepted) and the electrophile "nitronium ion / NO_2^+" (WCH15 Jan 2024).
- Temperature for nitrating methylbenzene: "below 30 °C ... to prevent further substitutions" (methylbenzene's ring is more reactive than benzene's). "Reagents decompose" is ignored (WCH15 Jun 2022).
- Sulfonation: "(concentrated) sulfuric acid" or "fuming sulfuric acid / oleum / $H_2S_2O_7$"; the product has SO_3H bonded through **S** (WCH15 Jun 2024).
- Friedel–Crafts methylation: "CH_3Cl / chloromethane" + "$AlCl_3$" (CH_3Br, CH_3I, $AlBr_3$, $FeBr_3$, $FeCl_3$ accepted). Chloroethane for a methyl group, or "a halogenoalkane" with no name, is not accepted (WCH15 Jan 2025).
- Sooty flame "indicates an arene / high C : H ratio" (WCH15 Jun 2022).

Often asked as: reagents and conditions for a step in a synthesis, the product of a given reagent, name the electrophile, why the temperature is controlled.

COMMON MISTAKE

Leaving out "concentrated" for the sulfuric acid in nitration (Jan 2024 report).

Choosing chloroethane when the product needs a methyl group: count the carbons (Jan 2025 report).

Calling bromination "Friedel–Crafts". Friedel–Crafts means alkylation or acylation (Jun 2023 report).

Drawing the sulfonic acid group joined through O, or on the wrong carbon (Jun 2024 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Substituted rings (methylbenzene, methoxybenzene)**: they react faster than benzene and substitute mainly at the **2- and 4-positions**. The 4-product is often the major one because a group next to the substituent is **sterically hindered** (WCH15 Jan 2026, acylation of methoxybenzene).
- **Side-chain oxidation**: methylbenzene heated under reflux with potassium manganate(VII), then acidified, gives **benzoic acid** (WCH15 Jan 2023).
- **Benzoic acid from benzene**: $Br_2/FeBr_3$ → bromobenzene; Mg in dry ether → C_6H_5MgBr; CO_2 then dilute acid → C_6H_5COOH (WCH15 Jan 2022, Oct 2023).
- **Limiting reagent** in a Friedel–Crafts prep: methoxybenzene 5.25 ÷ 108 = 0.0486 mol against ethanoyl chloride 4.50 ÷ 78.5 = 0.0573 mol, so methoxybenzene is limiting (WCH15 Jan 2026).

SPEC 18.5 Electrophilic substitution mechanisms

WHAT THIS SPEC POINT ASKS

Understand the electrophilic substitution mechanism for halogenation, nitration and Friedel–Crafts reactions of benzene, including how each electrophile is generated.

The pattern: three stages

1. **Make the electrophile** (a positive ion, E^+) using the catalyst.
2. **Attack**: the π electrons of the ring attack E^+. A positive **intermediate** forms and the delocalised ring is partly broken.
3. **Restore the ring**: the C–H bond breaks, H^+ is lost, and the delocalised ring re-forms. The H^+ regenerates the catalyst.

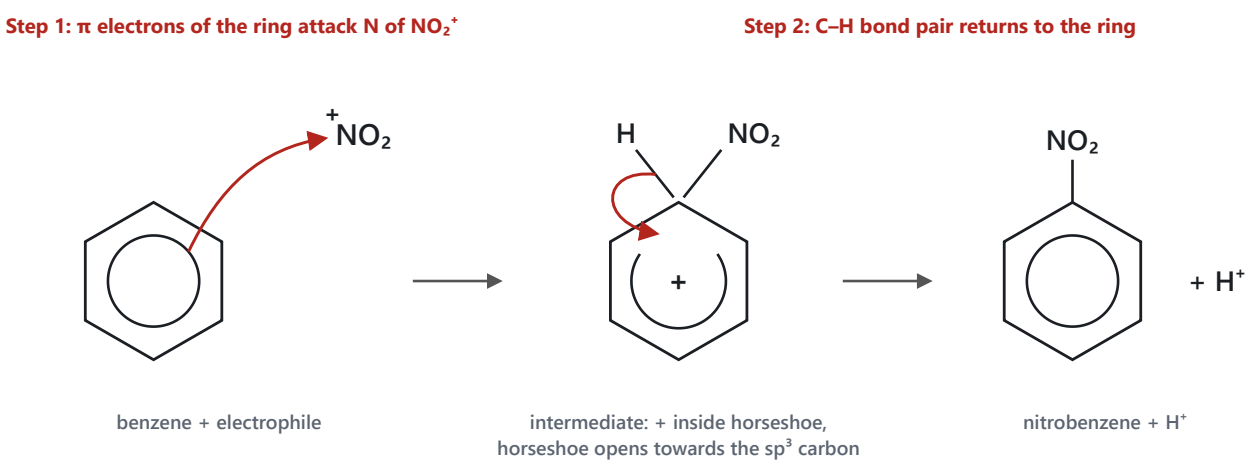
Stage 1: generating each electrophile

Reaction	Equation for the electrophile	Catalyst regenerated by
Nitration	$HNO_3 + 2H_2SO_4 \rightarrow \mathbf{NO_2^+} + 2HSO_4^- + H_3O^+$	$H^+ + HSO_4^- \rightarrow H_2SO_4$
Bromination	$Br_2 + FeBr_3 \rightarrow \mathbf{Br^+} + FeBr_4^-$	$H^+ + FeBr_4^- \rightarrow FeBr_3 + HBr$
Alkylation	$CH_3Cl + AlCl_3 \rightarrow \mathbf{CH_3^+} + AlCl_4^-$	$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$
Acylation	$CH_3COCl + AlCl_3 \rightarrow \mathbf{CH_3CO^+} + AlCl_4^-$	$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$

- **Nitration**: sulfuric acid is the stronger acid, so it protonates nitric acid. The protonated nitric acid ($H_2NO_3^+$) loses water to give the **nitronium ion, NO_2^+**. A simpler version is also accepted: $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$.
- **Halogenation**: $FeBr_3$ (or $AlCl_3$) has an **incomplete outer shell**, so it accepts a lone pair from one Br atom. This polarises and breaks Br–Br, leaving Br^+. Writing it as $Br\delta+–Br\delta-\cdots FeBr_3$ is also accepted.

- **Friedel–Crafts**: $AlCl_3$ takes the chlorine (with its bonding pair) from the C–Cl bond. That leaves a positive carbon: a **carbocation** R^+ or an **acylium ion** RCO^+ (the + is on the carbonyl carbon).

Stages 2 and 3: the mechanism with curly arrows



Nitration. For the other reactions, replace NO_2^+ with Br^+, CH_3^+ or CH_3CO^+. Everything else is the same.

Step 1: attack on the electrophile

1. Draw a curly arrow **starting inside the hexagon** (on or within the circle) and **ending at the positive atom** of the electrophile: the **N** of NO_2^+, the **Br** of Br^+, the **C** of CH_3^+ or CH_3CO^+.
2. Two of the π electrons form a new bond from one ring carbon to E.

The intermediate

1. The carbon now bonded to both **H and E** is **sp^3 (tetrahedral)**, so it is no longer part of the delocalised system. Draw both atoms on it.
2. The other five carbons share **four π electrons** and a **positive charge**. Draw a **horseshoe** (a partial circle) over these five carbons, with its **gap facing the tetrahedral carbon**, and put the **+ inside the horseshoe**.

Step 2: loss of H^+

1. Draw a curly arrow **from the C–H bond** (the middle of the bond, not the H atom) **back into the ring**. The bonding pair restores the full delocalised ring.
2. Products: the substituted arene (e.g. nitrobenzene) and **H^+**. The H^+ then regenerates the catalyst (table above).

Why substitution, not addition?

The intermediate could gain a negative ion (addition), but then the product would have lost its delocalised ring. **Losing H^+ restores the delocalised ring**, with its extra stability of about 150 kJ mol^{-1}, so substitution wins.

EXAM TIP

Mechanism checklist (each item is a marking point):

- Arrow 1 starts **inside the hexagon**, not on the edge or outside it.
- Arrow 1 ends on the **positive atom** (N of NO_2^+, C of CH_3^+ or CH_3CO^+), never on an O or on a lone pair.
- Electrophile shown with its **+ charge**.
- Horseshoe covers **at least three carbons** (ideally five), **open towards the sp³ carbon**, **+ inside**.
- Both H and E on the sp³ carbon.
- Arrow 2 starts on the **C–H bond** and goes into the ring. Use a **full (double-headed) arrow**: both electrons move.
- **H^+** as a product, then the catalyst-regeneration equation.

MARK-SCHEME WORDING

- "curly arrow from on or within the circle / hexagon to the positively charged atom (N of NO_2^+; C of the acylium ion)";
- "intermediate ... horseshoe covering at least 3 carbon atoms and facing the tetrahedral carbon, with some part of the positive charge within the horseshoe";
- "curly arrow from C–H bond to anywhere in the ring reforming the delocalised structure";
- electrophile equation, e.g. "$CH_3COCl + AlCl_3 \rightarrow CH_3C^+O + AlCl_4^-$", and regeneration "$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$".
- Not awarded: an arrow starting on or outside the hexagon; a missing + on the electrophile; an arrow to the O of NO_2^+; dotted C–H bonds; substituents on the wrong carbon.
- Error-spotting (WCH15 Jun 2022): the arrow drawn from CH_3^+ to the ring was "the wrong direction" (the ring is electron-rich, CH_3^+ is the electrophile with no lone pair). A single-headed arrow from the C–H bond is wrong because both electrons move (a half arrow means a radical). "Justify" needs the reason, not just the circled error.

Often asked as: the full 4–5 mark mechanism for nitration, bromination or acylation, nearly every series; find and fix the errors in a drawn mechanism; the equation for making NO_2^+.

COMMON MISTAKE

Starting arrow 1 from the electrophile, or aiming it at the O of NO_2^+ (Jan 2023, Jan 2025 reports).

The horseshoe's gap facing the wrong way, or the + drawn outside it, or left on the NO_2 group (Oct 2020, Jun 2024 reports).

Starting arrow 2 from the H atom instead of the C–H bond (Jan 2025 report).

Adding an extra carbon to the electrophile in Friedel–Crafts, e.g. writing $CH_3CH_2CO^+$ for ethanoyl chloride (Oct 2024 report).

Forgetting to show the catalyst being regenerated.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Unfamiliar electrophiles**: CO + HCl with $AlCl_3$ gives the **formyl cation HCO^+**, which adds a –CHO group to the ring (WCH15 Oct 2022). Treat any new electrophile exactly like NO_2^+: arrow to its positive atom.
- A **carbocation from an alkene** can also be the electrophile, e.g. when $AlCl_3$ and a chloro-acid are used (WCH15 Jan 2021).
- **Substituted rings**: put the substituent and E on the correct carbons in the intermediate. For phenol, the new group goes at position 2 or 4 (WCH15 Jan 2023).

SPEC 18.6 Phenol and bromine water

WHAT THIS SPEC POINT ASKS

Understand the reaction of phenol with bromine water, and why phenol reacts much more easily than benzene.

The reaction

$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$

- **Conditions**: bromine **water**, **room temperature**, **no catalyst**.
- **Observations**: the bromine water **decolourises** (orange → colourless), and a **white precipitate** of **2,4,6-tribromophenol** forms. It smells of antiseptic.
- Three H atoms are substituted, at **positions 2, 4 and 6** (the carbons next to and opposite the –OH). Carbon 1 carries the OH.

Why phenol reacts more easily than benzene

1. One **lone pair on the oxygen atom** of the –OH group sits in a p orbital that **overlaps with the π system** of the ring.
2. The lone pair becomes **partly delocalised into the ring**, which **increases the electron density** of the ring (most at positions 2, 4 and 6).
3. The electron-rich ring can **polarise** Br_2 molecules by itself, so no halogen carrier is needed, and it **attracts electrophiles more strongly**.
4. So phenol undergoes **electrophilic substitution** much more readily than benzene, and substitutes three times.

	Benzene	Phenol
Reagent	Br_2 (liquid)	bromine water

	Benzene	Phenol
Catalyst	$FeBr_3$ needed	none
Product	bromobenzene (one Br)	2,4,6-tribromophenol (three Br)
Observation	slow; HBr fumes	instant decolourisation, white precipitate

MARK-SCHEME WORDING

- "lone pair of electrons on the **oxygen**"; "overlaps / interacts with the (π) electrons of the ring"; "**increasing** the electron density of the ring"; "so phenol is more susceptible to electrophilic attack" (WCH15 Jun 2024).
- Product: "2,4,6-tribromophenol" (Kekulé drawing accepted); equation "$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$".
- Ignored: "increases the reactivity of the ring", "phenol is a nucleophile", "milder conditions" alone, "quicker / easier". Not accepted: "charge density increases", "the ring becomes more electronegative", "electrophilic addition".

Often asked as: name or draw the product (1–2 marks), explain the greater reactivity (2–3 marks), or the 6-mark comparison of cyclohexene, benzene and phenol (WCH15 Oct 2022, Oct 2023, Jan 2025).

COMMON MISTAKE

Write "a lone pair on the **oxygen atom**", not "a lone pair on the OH": one mark scheme refused that wording (WCH15 Jan 2025).

Stating "the lone pair" without saying **where** it is, or not linking the lone pair to **higher electron density in the ring** (Jan 2024, Jan 2023 reports).

Writing "long pair". Drawing the Br atoms at positions 2, 3 and 4.

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- **Phenylamine** does the same thing with bromine water: the lone pair on N activates the ring, giving a white precipitate of 2,4,6-tribromophenylamine.
- **Nitration of phenol** needs only **dilute** nitric acid at room temperature and gives a mixture of **2-nitrophenol and 4-nitrophenol** (WCH15 Jan 2023). Draw the NO_2 on carbon 2 or 4, never on the carbon carrying the OH.
- **Methoxybenzene** ($C_6H_5OCH_3$) is activated in the same way by the O lone pair. The spec suggests it, or methylbenzene, for practical work instead of the more hazardous benzene.

Quick checks

QUICK CHECK

Which observation is the best evidence that benzene's six C–C bonds are identical?

- benzene burns with a smoky flame
- X-ray diffraction shows all six C–C bonds are 0.140 nm long
- benzene is a liquid at room temperature
- benzene reacts with hydrogen over nickel

Kekulé's structure would have three short C=C and three long C–C bonds. One intermediate length shows the π electrons are spread evenly round the ring.

QUICK CHECK

The enthalpy of hydrogenation of cyclohexene is −120 kJ mol^{-1} and of benzene −208 kJ mol^{-1}. What is the delocalisation energy of benzene?

- 88 kJ mol^{-1}
- 208 kJ mol^{-1}
- 152 kJ mol^{-1}
- 328 kJ mol^{-1}

Three C=C bonds would release 3 × 120 = 360 kJ mol^{-1}; benzene releases 208, so it is 360 − 208 = 152 kJ mol^{-1} more stable. 88 is 208 − 120, which compares with only one double bond.

QUICK CHECK

Why does benzene undergo substitution rather than addition with electrophiles?

- benzene has no double bonds at all
- electrophiles cannot attack benzene
- substitution keeps the stable delocalised π system, which addition would destroy
- addition products are too unstable to isolate

Losing H^+ at the end restores the full ring of delocalised electrons and its delocalisation energy.

QUICK CHECK

What is the electrophile in the nitration of benzene, and how is it formed?

- NO_3^-, from nitric acid losing H^+
- NO_2^+, from $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$
- H^+, from sulfuric acid
- NO_2, a radical made by heating

Sulfuric acid, the stronger acid, protonates nitric acid, which then loses water to give the nitronium ion. Sulfuric acid is regenerated at the end.

QUICK CHECK

Which reagents would convert benzene into phenylethanone, $C_6H_5COCH_3$?

- ethanol and concentrated sulfuric acid
- chloroethane and $AlCl_3$
- ethanoyl chloride and anhydrous $AlCl_3$
- ethanoic acid and $FeBr_3$

Friedel–Crafts acylation: $AlCl_3$ forms the acylium ion CH_3CO^+ from ethanoyl chloride. Chloroethane would give ethylbenzene by alkylation.

QUICK CHECK

How can you distinguish phenol from ethanoic acid?

- add sodium: only phenol fizzes
- add sodium hydroxide: only phenol reacts
- add sodium carbonate: only ethanoic acid fizzes
- measure the pH: both are neutral

Both react with sodium and with sodium hydroxide, but phenol is too weak an acid to release CO_2 from a carbonate.

QUICK CHECK

Why does phenol react with bromine water without a catalyst, while benzene does not?

- phenol is a stronger acid than benzene
- a lone pair on oxygen is delocalised into the ring, raising its electron density enough to polarise Br_2
- the OH group is an electrophile
- phenol has a double bond outside the ring

The extra electron density makes the ring so attractive to electrophiles that Br_2 is polarised directly, and substitution happens at three positions to give 2,4,6-tribromophenol.

QUICK CHECK

In the mechanism of electrophilic substitution, what does the "horseshoe" in the intermediate show?

- the electrophile is attached to every carbon
- the remaining delocalised electrons and the positive charge are spread over five carbons
- the ring has opened
- the hydrogen has already been lost

Two of the six π electrons have formed the bond to the electrophile, so the other four, with the positive charge, are delocalised over the remaining five carbons. The sixth carbon carries both H and E.